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Bio-Inspired Cascade Photocatalysis on Fe Single-Atom Carbon Nitride Upcycles Plastic Wastes for Effective Acetic Acid Production

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ABSTRACT

Plastic imposes a critical threat to the environment, ecosystems, and human health because of the low utilization efficiency of plastics. Here, we demonstrate a sustainable, highly efficient cascade photocatalysis for upcycle plastics to value-added acetic acid using Fe single-atom catalysts (Fe@C_3N_4 SAC) at ambient conditions. Inspired by Phanerochaete chrysosporium microbial, the defective Fe@C_3N_4 SAC acts as a bifunctional cascade photocatalyst for both Fenton-like and CO_2 reduction reactions. During the reaction, hydroxyl radicals (*OH) form and subsequently oxidize plastics into CO_2 intermediates. These CO_2 intermediates are then photo-reduced to CH_3COOH on the same catalyst via cascade photocatalysis. The mechanism is confirmed by in situ multimodal microscopy and spectroscopies, with density functional theory calculations. A state-of-art CH_3COOH yield of 63.8 mg h^{-1} g_{cat}^{-1} from PVC, 12.7 mg h^{-1} g_{cat}^{-1} from PE, 5.4 mg h^{-1} g_{cat}^{-1} from PET, and 5.3 mg h^{-1} g_{cat}^{-1} from PP are directly obtained under AM1.5G solar irradiation and further validated under real sunlight (≈0.6 sun), achieving 5.6 mg h^{-1} g_{cat}^{-1} from PET, using low-cost Fe@C_3N_4 SAC in a sealed reactor by enhancing the photon transport and utilization efficiency. The techno-economic analysis shows it is promising to practically mitigate plastic based on broader social welfare assessments.

1 | Introduction

Since the 1950s, the plastics industry has rapidly developed and contributed significantly to technological and economic advances [1, 2]. Plastic wastes possess a very long self-degradation cycle in nature (ca. 250–500 years) due to their strong chemical inertness, causing the worldwide "white pollution" problem [3]. Extensive research shows that all major ecosystems on Earth have been

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contaminated by plastic wastes, including microplastics, raising concerns about its potential threats to terrestrial and marine life and human health [4, 5]. In particular, the appearance of plastic wastes in our food and drinking water supplies (e.g., seafood, tea, and vegetables), is a significant worry [6, 7]. Microplastics have even been found in human placenta [8] and infant formula [9]. The COVID-19 pandemic has further intensified pressure on an already out-of-control global plastic waste problem by significantly increasing the amount of used disposable plastic medical supplies, including syringes, gloves, and other personal protective equipment [10]. Therefore, it is urgent to develop efficient routes for upcycling plastic waste.

Landfill and incineration are the most used traditional commercialized methods for plastic waste treatment [11–14]. Although these methods seem to work, they can't eliminate the plastic wastes and may even cause serious environmental issues, such as plastic debris contamination of groundwater from landfills, and release of toxic and greenhouse (CO_2) gases from incineration or thermo-chemical methods [15]. Mechanical recycling is another traditional method, which can recycle plastic wastes by mechanically grinding the plastics into secondary raw materials, but this approach is better considered a downgrading recycle method, and still leads to plastic waste production. Besides, thermo-catalytic pyrolysis of plastic wastes into valuable carbonaceous fuels [16, 17] and pyrolysis of plastics at extreme high temperature [18] have also been reported, but the harsh experimental conditions (high temperature, high energy input, high pressure, inert atmosphere, and costly metal complex catalysts) limits their practical applications. Recently, Uekert et al. reported a two-step photo-reforming strategy to transform alkaline heat pretreated hydrolysed plastics products into a variety of small carbonaceous molecules under room temperature, and producing H_2 at the same time [19–21]. However, the alkaline heat treatment cost extra energy, and the strongly alkaline solution used in the hydrolytic process may inevitably cause adverse environmental effects. This two-step photo-reforming strategy causes plastic degradation by alkali hydrolysis rather than one-step photocatalysis processes. The solar energy is just used to transform the hydrolyzed plastics products, rather than the plastic itself. Thus, the energy efficiency of this two-step process is low. Besides, the alkaline hydrolysis is primarily effective for plastic containing ester bonds, such as PET, and shows limited degradation capability for other polymers like PP. To better utilize the solar energy, avoid large amounts of environmentally unfriendly alkaline solutions, and degrade various types of plastics, it needs to develop one-step process to upcycling plastic using solar energy. Jiao et al. developed a Nb_2O_5 atomic layer catalyst photocatalyst to transform plastic wastes into value-added products [22]. But the very low production yield and high cost of Nb_2O_5 atomic layers significantly limits the scaled-up industrial application of this method. Recent efforts have provided promising directions for addressing these challenges. Sharma et al. summarized the potential of non-noble-metal-based heterogeneous catalysts for hydrogenation reactions, which offer detailed insights into alternative hydrogen sources under mild conditions [23]. Similarly, Sajwan et al. summarized innovative photo-, electro-, and photoelectrocatalytic approaches for upcycling plastic waste with hydrogen production, contributing to the advancement of circular economy principles through more sustainable methods [24]. Given this, it is highly urgent to develop innovative solutions to convert plastic wastes into value-added chemicals with high production yield under ambient conditions using a low-cost photocatalyst.

In nature, microorganisms such as fungi provide many degradation methods for polymerized wood such as lignin and cellulose. This is achieved mainly by the use of peroxidases to break the bonds between various lignin units [25]. Among them, Phanerochaete chrysosporium is able to produce lignin-degrading enzymes such as lignin peroxidase (LiP) and manganese peroxidase (MnP) [26]. LiP and MnP will attack lignin nonspecifically triggered by H_2O_2, thereby achieve primary degradation of lignin [27, 28]. During this Fenton-like process, no CO_2 was released from Phanerochaete chrysosporium.

Inspired by this bacteria, here, we designed a low-cost, and highly efficient Fe single-atom catalyst (Fe@C_3N_4 SAC) that serves as a cascade photocatalyst by coupling the Fenton and CO_2 reduction reactions. This photocatalyst can not only reduce material costs but also produces value-added acetic acid from various plastic wastes, such as PET, PP, PE and PVC. Similar as bacteria Phanerochaete chrysosporium, H_2O_2 is first photocatalytic converted into a hydroxyl radical (*OH) over an Fe single atom active site on Fe@C_3N_4 SAC. Subsequently, the produced hydroxyl radical oxidizes plastics into CO_2 intermediates, which are then further reduced into acetic acid on the Fe@C_3N_4 SAC. This reaction mechanism was confirmed by multimodal in situ synchrotron X-ray absorption spectroscopy (XAS), in situ Fourier transform infrared (FTIR) spectroscopy, in situ electron paramagnetic resonance (EPR), in situ UV–vis-absorption spectroscopy, Raman spectroscopy, and aberration-corrected scanning transmission electron microscopy, nuclear magnetic resonance spectroscopy, and density functional theory calculations. The Fe@C_3N_4 SAC can avoid the common problem of metal leaching and massive sludge generation by the traditional Fenton reaction, since the extremely low amount of Fe atoms are well bonded on the C_3N_4 support framework. With this rational design, the acetic acid production in our one-step photocatalytic system reached a landmark activity of 10.5 mg h^{-1} $g_{cat}{}^{-1}$ from PE, 2.96 mg h^{-1} $g_{cat}{}^{-1}$ from PET, and 1.7 mg h^{-1} $g_{cat}{}^{-1}$ from PP under AM1.5G irradiation. Additionally, our work introduced several innovative aspects, such as innovations in reactor design that contribute to improved overall efficiency and operational sustainability. The production yield can be greatly boosted by sealing the reactor with Aluminum (Al) foil to enhance photon transport and utilization. Acetic acid production in the same photocatalytic system with the reactor covered by Al foil is 63.8 mg h^{-1} $g_{cat}{}^{-1}$ from PVC, 12.7 mg h^{-1} $g_{cat}{}^{-1}$ from PE, 5.4 mg h^{-1} $g_{cat}{}^{-1}$ from PET, and 5.3 mg h^{-1} $g_{cat}{}^{-1}$ from PP. Importantly, under real sunlight illumination (≈0.6 sun), a consistent yield of 5.6 mg h^{-1} $g_{cat}{}^{-1}$ was also achieved for PET. The yield of acetic acid is about 5.5 mg h^{-1} $g_{cat}{}^{-1}$ from mixed plastics, containing 33% PE, 34% PET and 33% PP. This represents the state-of-the-art in one-step photocatalytic conversion of diverse plastic wastes (PET, PE, PP, and PVC) (Table S1), demonstrating substantial advancements over prior methodologies in terms of economic viability and catalytic efficiency. Besides, unlike alkaline solution, H_2O_2 is an environmentally benign reagent. It is decomposed into water and oxygen, producing no harmful byproducts to the environment. The integration of multiple innovative elements distinctly sets this work apart from previous studies and underscores its significant contribution to the field.

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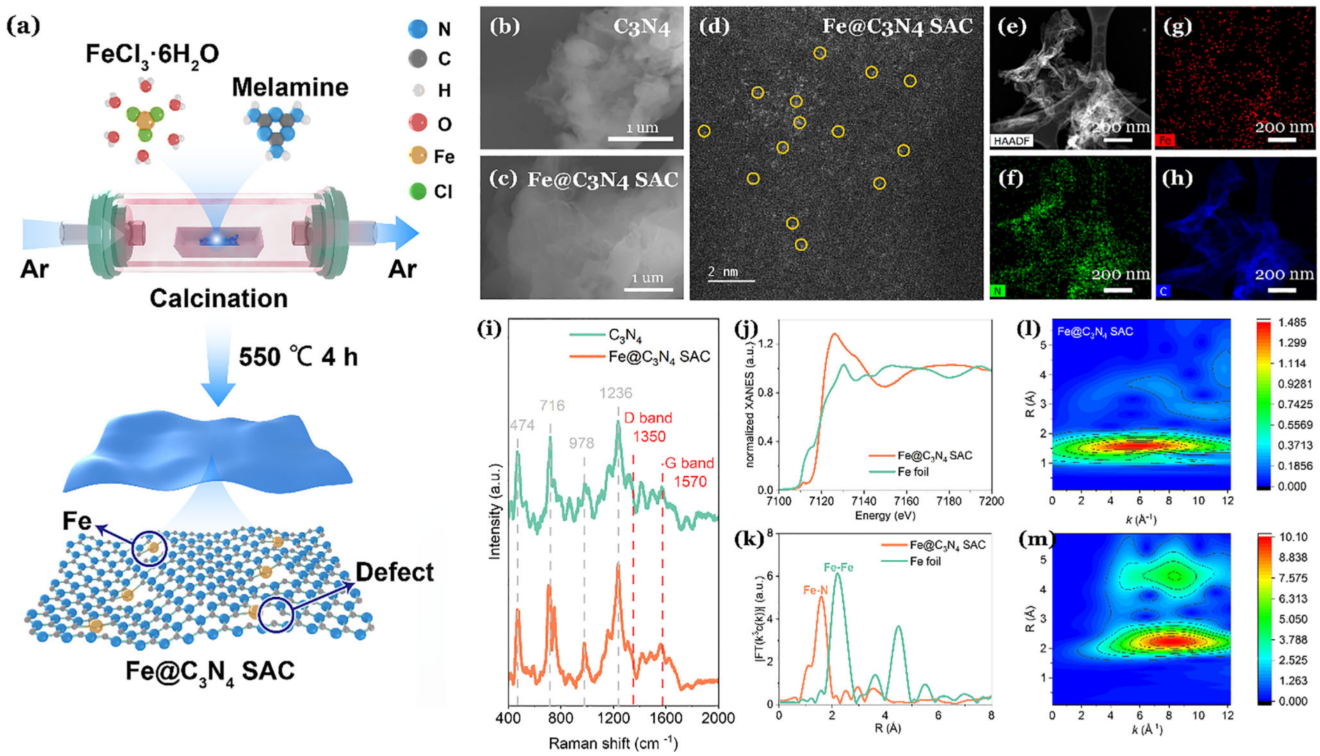

FIGURE 1 | Synthesis and Ex situ Materials Characterizations using scanning electron microscopy (SEM), aberration-corrected scanning transmission electron microscopy (AC-STEM) characterization, ex situ spectroscopy characterization and hard X-ray absorption spectroscopy characterization. (a) Preparation of $Fe@C_3N_4$ SAC. (b) SEM image of C_3N_4. (c) SEM image of $Fe@C_3N_4$ SAC. (d) Aberration corrected High-angle annular dark-field scanning transmission electron microscopy (AC-HAADF-STEM) image of $Fe@C_3N_4$ SAC; the Fe single atoms are marked by yellow circles. (e)–(h) scanning transmission electron microscopy energy dispersive X-ray spectroscopy (STEM-EDX) elemental mapping images of $Fe@C_3N_4$ SAC. (i) Raman spectroscopy of C_3N_4 and $Fe@C_3N_4$ SAC. The Raman spectra were recorded with an exciting laser at 785 nm. Synchrotron X-ray spectroscopy characterization of Fe@ C_3N_4 photocatalyst from Advanced Photon Source. (j) Fe K-edge X-ray absorption near-edge structure (XANES) spectra of $Fe@C_3N_4$ SAC and Fe foil. (k) Fourier-transformed (FT) curves at R space for $Fe@C_3N_4$ SAC and Fe foil. (l-m) Wavelet transform (WT) of EXAFS at Fe K-edge for $Fe@C_3N_4$ SAC and Fe foil as a reference.

1.1 | Synthesis and Ex-Situ Characterization of the $Fe@C_3N_4$ SAC Photocatalyst

The synthesis procedures of graphitic carbon nitride (C_3N_4) and $Fe@C_3N_4$ SAC are illustrated (Figure 1a, Experimental Section). Melamine was thoroughly mixed with a certain amount of iron salt and calcined in a tube furnace to finally obtain the photocatalyst. The scanning electron microscope (SEM) image (Figure 1b) shows the morphology of the C_3N_4 nanosheet. The crumpled nanosheets are highly curved and able to stack on top of each other like petals. The loose morphology allows the materials to have pores of a few microns in diameter, which facilitates adsorption and contact between the active molecules and the organic matter [29, 30]. $Fe@C_3N_4$ SAC shows a similar structure (Figure 1c), but the Fe single atoms cannot be observed in SEM due to inadequate spatial resolution. The maintenance of morphology reveals that Fe incorporation into C_3N_4 does not change the original lamellar structure [31]. To reveal the distribution of Fe single atoms in the catalyst, probe aberration-corrected high-angle annular dark field scanning transmission electron microscopy (AC-HAADF-STEM) was carried out. According to the z-contrast, the bright spots (examples labelled with yellow circles in Figure 1d) represent Fe single atoms, which are uniformly dispersed on the C_3N_4 support framework. This indicates that Fe single atoms were successfully introduced into the C_3N_4 structure (Figure 1d). To ascertain the structural stability of the Fe single atom sites under operating conditions, the catalyst was collected after 12-h irradiation and subjected to identical microscopic analysis. As evidenced by the AC-HAADF-STEM image (Figure S1), the characteristic bright dots indicative of isolated Fe single atoms remained abundantly dispersed on the C_3N_4 support, with no observable formation of nanoparticles or clusters. The corresponding EDX elemental mapping (Figure S1) further confirmed the homogeneous distribution of Fe, N, and C, mirroring the pattern of the fresh catalyst. These results provide direct visual evidence that the atomic dispersion of Fe is robustly preserved throughout the catalytic process, which is a cornerstone of the catalyst's enduring performance. Energy dispersive X-ray spectroscopy (EDX) elemental mapping results (Figure 1e–h) further confirm homogenous distribution of Fe single atoms on the C_3N_4 support framework. Inductively coupled plasma mass spectrometry (ICP-MS) analysis (Experimental Section) indicates that the Fe content in $Fe@C_3N_4$ SAC was 0.5 wt.%. This result explains why the signal of Fe single atoms is relatively weak in EDX mapping (Figure 1g). Similarly, the post-reaction samples were subjected to ICP-MS analysis. The result shows an iron mass fraction of 0.5 wt.%, indicating there is no leaching of iron from the reaction.

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X-ray powder diffraction (XRD) analysis of C_3N_4 and Fe@C_3N_4 SAC was carried out (Figure S2). Two diffraction peaks appeared at 13.3° and 27.4° in C_3N_4, which are referred to as the standard card (JCPDS 87–1526). As shown in Figure S2, the sharp peaks at 27.4° are formed by the π–π interlayer stacking reflection with an interplanar separation of 0.34 nm, indexed as the (002) diffraction peak [32, 33]. The broad peaks located at 13.3°, indexed as (100), are generated due to the in-plane structural packing motif, representing an interplanar distance of 0.68 nm [33, 34]. Compared to C_3N_4, Fe@C_3N_4 SAC shows a decreased intensity of diffraction peaks of the (100) and (002) planes. This is due to the formation of Fe-N_x coordination bonds between iron and nitrogen atoms during calcination, which ruins the repetitive arrangement of the basic units in C_3N_4 [35]. The results indicate that Fe species are sufficiently embedded in the C_3N_4 skeleton. To evaluate the structural integrity of the C_3N_4 and Fe@C_3N_4 SAC framework after reaction, the XRD pattern of the spent catalyst was compared with that of the fresh sample (Figure S2). It was observed that the intensity of the (100) and (002) peaks decreased, although their positions remained unchanged. This indicates that the fundamental layered structure of C_3N_4 was preserved. Furthermore, the embedding of Fe species restructures the material into a more porous framework, as confirmed by N_2 adsorption-desorption distribution curves. The Fe@C_3N_4 SAC exhibits a specific surface area of 17.8 m^2 g^{-1} and an average pore width of 8.7 nm, characteristic of a mesoporous material (Figures S3 and S4). This altered texture provides a favorable environment for the adsorption and diffusion of reactant molecules, thereby contributing to the enhanced catalytic efficiency observed in the plastic upcycling process.

The chemical valence states and elemental compositions were studied by X-ray photoelectron spectroscopy (XPS). High resolution of C 1s XPS spectrum is depicted (Figure S5). The peak is deconvoluted into three peaks of C—C, C—N—C, and C—O bonds, which are located at 285.1, 288.2, and 289.2 eV, respectively (Figure S5) [36–38]. The C 1s spectrum of C_3N_4 and Fe@C_3N_4 SAC are similar. However, the peak area for Fe@C_3N_4 SAC corresponding to the $(N)_2$—C═N groups at 288.2 eV is larger (Figure S5) because Fe-N_x bonds formed and affected the chemical environment of the adjacent C atoms in the original structure [35]. Given that it is easy for carbon peaks to be contaminated by the background environment, the nitrogen spectra are key when considering the bonding structure [36]. The characteristic N 1s peaks could be clearly observed (Figure S6). Three typical N 1s core levels are located at 398.60, 399.81, and 401.11 eV (Figure S6a). These peaks correspond to sp^2 C—N═C, sp^3 N—C_3 and C—N—H (—NH_2 or ═NH) [36–39] and the fractional abundances of these bonds are 0.28, 0.55 and 0.17, respectively (Figure S6a). It is noted that N-Fe bonds are also located at 398.8 ± 0.5 eV, and are difficult to distinguish from C-N = C bonds [40, 41]. A similar characteristic peak can also be observed in the N 1s high-resolution spectrum of Fe@C_3N_4 SAC, which further indicates the successful embedding of Fe species into the C_3N_4 lattice (Figure S6b). From Fe 2p spectra in Fe@C_3N_4 SAC, two main broad peaks at about 710 and 724 eV are observed, which can be attributed to typical Fe $2p_{3/2}$ and Fe $2p_{1/2}$ signals [42]. These two main peaks could be separated into sub-peaks (Figure S7a). The absence of a metallic Fe peak indicates that Fe atoms successfully connect to C_3N_4, which is consistent with the HAADF-STEM imaging (Figure 1d). The Fe 2p spectra of Fe@C_3N_4 SAC after light irradiation (Figure S7) reveals a noticeable increase in the proportion of satellite peaks. This observation may be attributed to changes in the coordination environment and alterations in the electronic structure of the single-atom iron sites. During the light irradiation process, photogenerated electrons reduce Fe^{3+} to Fe^{2+}, leading to a decrease in Fe^{3+} (Figure S7). However, the Fe^{2+} species are rapidly consumed in subsequent Fenton reactions. As a result, the overall concentration of Fe^{2+} does not exhibit a significant change, despite the dynamic redox cycle facilitated by light exposure (Figure S7). The use of Fe single atoms offers several advantages, including dural active sites for both Fenton reaction and CO_2 conversion reaction, higher atomic utilization efficiency, stable dispersion on the C_3N_4 support, and enhanced catalytic activity due to the unique electronic and coordination environment of the single atoms.

To investigate the surface defects of the synthesized catalysts, Raman spectroscopy was carried out. The peaks located at 1350 and 1570 cm^{-1}, respectively, are attributed to the D peak and the G peak (red dashed line in Figure 1i) [43]. It is noted that the I_D: I_G value of Fe@C_3N_4 SAC is larger than that of C_3N_4, indicating an increased defective degree. This may be due to the breaking of N═C—N bonds by introducing Fe single atoms on C_3N_4 [44]. In addition, there are other distinct peaks (Figure 1i). The vibrational peaks at 474 and 716 cm^{-1} (Figure 1i) are associated with characteristic vibrations of s-triazine ring and C═N, respectively [45]. Other dominant peaks at 978 and 1236 cm^{-1} (Figure 1i) are separately derived from breathing modes of symmetric N in heptazine units and s-triazine [45, 46].

UV–vis absorption spectroscopy was carried out to analyze the optical properties of C_3N_4 and Fe@C_3N_4 SAC (Figure S8). Both C_3N_4 and Fe@C_3N_4 SAC have the same energy absorption edge in the ultraviolet and visible light region (Figure S8). As an excellent light absorption material, C_3N_4 has an intrinsic absorption edge at about 460 nm (≈2.7 eV) [47]. The introduction of Fe single atoms does not change the absorption edge of C_3N_4.

To better understand electronic structures and the coordination environment of the Fe atoms, the hard X-ray absorption spectroscopies were measured at the Advanced Photon Source, Argonne National Laboratory. The Fe K-edge X-ray absorption near-edge structure (XANES) spectra of Fe@C_3N_4 SAC is presented with Fe foil as a reference (Figure 1j; Figure S9a). Figure 1j shows that the absorption edge is higher than that of the Fe foil, demonstrating that the valence of Fe single atoms in Fe@C_3N_4 SAC is higher than metallic Fe [48]. Moreover, a pre-edge peak around 7114 eV can be attributed to the Fe-N_4 structure, consistent with previous reports [49, 50]. The k^3-weighted Fourier transformation of Fe K-edge from extended X-ray absorption fine structure (EXAFS) spectra for Fe@C_3N_4 SAC and Fe foil is shown to further verify the accurate coordination information (Figure 1k; Figure S9b). As depicted, the spectrum is dominated by a single intense peak at ≈1.6 Å, corresponding to the first coordination shell of Fe. Critically, the position and line shape of this peak show a remarkable similarity to the characteristic Fe—N peak of the iron phthalocyanine (FePc) standard, a molecule with a well-defined Fe—N_4 planar structure [51, 52]. This close match strongly indicates that the Fe center in our catalyst is coordinated by four N atoms, forming an analogous Fe—N_4 moiety. Furthermore, the absence of any significant contribution at higher R-space (e.g.,

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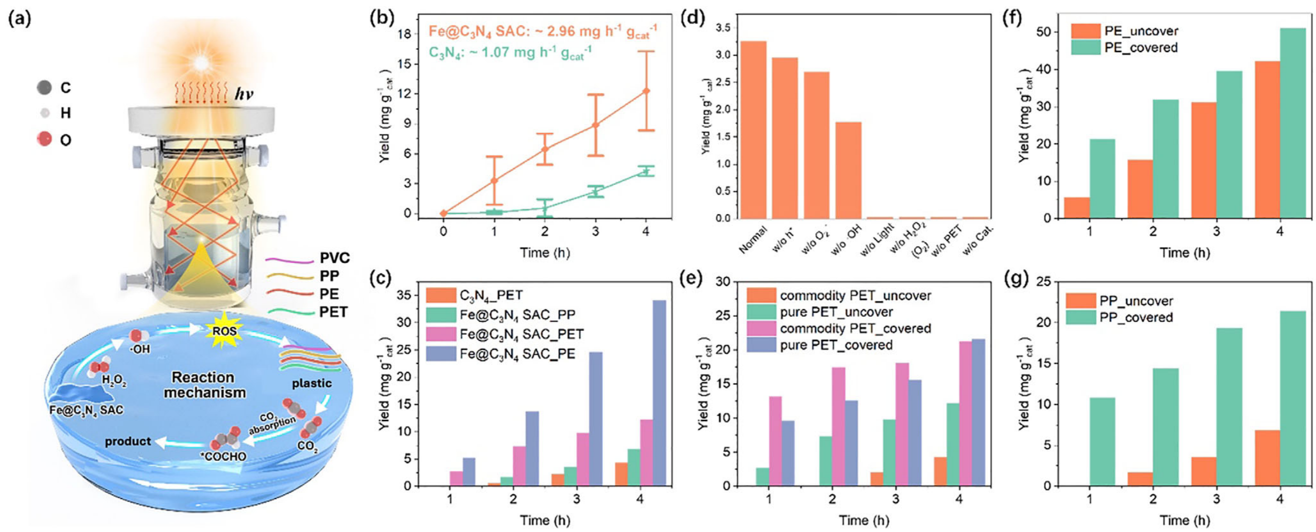

FIGURE 2 | Performance of plastic upcycling. (a) Experimental set up and mechanism of the plastic upcycling process. (b) Yield of acetic acid during the photocatalytic-induced degradation of pure PET with C_3N_4 (green line) or $Fe@C_3N_4$ SAC (orange line). The error bars in (b) represent the standard deviations of three independent measurements of the same sample. (c) Plastic upcycling of PET, PE, and PP using an uncovered reactor with $Fe@C_3N_4$ SAC or C_3N_4. (d) Photocatalytic activation of $Fe@C_3N_4$ SAC for PET degradation in the presence of different scavengers. (e) Effect of light trapping in the reactor for commodity PET from DuPont with 30% glass fibers and pure PET upcycling to acetic acid using the $Fe@C_3N_4$ SAC photocatalyst. (f) Effect of light trapping in the reactor for PE upcycling to acetic acid using the $Fe@C_3N_4$ SAC photocatalyst. (g) Effect of light trapping in the reactor for PP upcycling to acetic acid using the $Fe@C_3N_4$ SAC photocatalyst.

the Fe-Fe bond peak at ≈2.2 Å observed in the Fe foil reference) verifies the exclusive presence of isolated Fe atoms without the formation of clusters or nanoparticles [50, 53]. The corresponding fitting results are presented (Figures S10–S13 and Tables S2 and S3). Wavelet transform (WT) analysis of $Fe@C_3N_4$ SAC shows one maximum intensity at 4 $Å^{-1}$, matching well with the Fe-N bond in the reference FePc (Figure 1l) [54]. Compared with the WT plots of Fe foil (Figure 1m), there is no metallic Fe-Fe scattering signal (≈8 $Å^{-1}$) detected [51, 55]. Combined with the absence diffraction peaks associated with the crystalline iron species (Figure S2), it is further confirmed that iron are present as single atoms that connect with N atoms and are fixed in the C_3N_4 framework, which is consistent with the AC-HAADF-STEM measurement (Figure 1d).

1.2 | Photocatalytic Performance of the $Fe@C_3N_4$ SAC Photocatalyst

The photocatalytic activities of $Fe@C_3N_4$ SAC and C_3N_4 are all evaluated by the upcycling of plastic, such as polyethylene terephthalate (PET), under simulated conditions of one sun radiation and atmospheric pressure (Figure 2a). In the process, plastic is converted to CO_2 and immediately produces *COCHO, which is eventually converted to acetic acid. PET, typical commercial plastics for drinking bottles, is known to exhibit relatively stable physical and chemical properties under visible light irradiation without the involvement of a photocatalyst. $Fe@C_3N_4$ SAC shows a great improvement on the production of acetic acid compared with C_3N_4 due to the existence of Fe active sites (Figure 2b). The yield of the $Fe@C_3N_4$ SAC system under simulated sunlight is ≈2.96 mg h^{-1} $g_{cat}{}^{-1}$. Importantly, this activity is maintained under natural sunlight (see experimental setup, Figure S14, ≈0.6 sun), with an average acetate production rate of 3.3 mg h^{-1} $g_{cat}{}^{-1}$ (Figure S15). Notably, the catalyst exhibits exceptional operational stability and even intriguing activation behavior. As demonstrated in the recyclability tests (Figure S16), the acetic acid yield progressively increased over three consecutive 12-h irradiation, surpassing the initial yield. This suggests that the catalytic system becomes more efficient with use, likely due to the improved accessibility of the Fe single atom sites under prolonged reaction conditions. By contrast, C_3N_4 has a low activity for degrading PET, which is about a third of the activity of $Fe@C_3N_4$ SAC. The combination of high initial activity, sustained performance under natural sunlight, and enhanced stability positions the $Fe@C_3N_4$ SAC as a superior catalyst for acetate production under mild conditions without the need for acid/base pretreatment or high light intensity (Table S1).

To verify whether CO_2 is an intermediate during the plastic upcycling photocatalytic reaction, we replaced the plastics with CO_2 gas while keeping all experimental conditions the same. When no plastic was added but pure CO_2 gas flowed into the reactor, acetic acid was still produced by the photocatalyst (Figure S17). This indicates that the plastic first undergoes a C─C bond cleavage process and oxidizes into CO_2 in the Fenton reaction, and then CO_2 is selectively converted to acetic acid by the CO_2 reduction reaction.

To directly monitor the H_2O_2 consumption kinetics and confirm its sufficiency in driving the cascade reactions, we tracked its concentration profile via a potassium permanganate titrimetric method (Figure S18). All chemicals were used in the same amount in the first hour in the $Fe@C_3N_4$ SAC/H_2O_2/Vis system (Figure 2b). In the system, the initial H_2O_2 concentration was ≈0.6 M (Figure S18). As the reaction progressed, the concentration decreased but remained at a relatively high level of ≈0.3 M even after 4 h (Figure S18). This persistent presence of H_2O_2,

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which was added only at the beginning without replenishment, provides direct experimental evidence for its excess amount throughout the reaction. This is further corroborated by the continued production of acetic acid observed over the same period (Figure S18), confirming that the *OH radical generation, and thus the plastic degradation, was not limited by H_2O_2 availability.

Only talks about different plastic waste mixture as feed.

Does not take into account the additives being used in production.

The conversion rate of PET was tested over 54 h and showed a conversion rate of 0.1 mg h^{-1} mg_{cat}^{-1} under the Fe@C_3N_4 SAC/H_2O_2/Vis system (Methods). Our system demonstrates promising plastic upcycling efficiency and is effective for diverse plastic wastes (i.e., PP, PE and PET) (Figure 2c). Furthermore, we also conducted experiments with mixed plastics, containing 33% PE, 34% PET and 33% PP. Notably, the yield of acetic acid is ≈5.5 mg h^{-1} g_{cat}^{-1} (Figure S19), which is essentially the arithmetic mean of the production rates obtained from the individual polymers. This result suggests the absence of significant synergistic or competitive effects during co-degradation, implying that the reactive oxygen species in our system non-preferentially attack the different polymer chains. This finding indicates that our system is not only capable of efficiently converting mixtures of plastic waste into acetic acid but is also robust against variations in plastic composition, a highly desirable feature for treating real-world, complex waste streams. Intriguingly, a higher yield of acetic acid was observed when using PE. One possible reason is that the smaller the particle size of the plastic, the greater the specific surface area in contact with the catalyst solution, and the more acetic acid is produced (Figure S20). In addition, compared to PET at the same size, PE yields more acetate products, which can be understood by considering a sequential oxidative-reductive conversion pathway. The superior performance of PE stems from its uniform hydrocarbon structure, which is efficiently and completely oxidized to provide a high flux of CO_2 for the productive reductive coupling step. In contrast, the complex structure of PET impedes this pathway: its aromatic terephthalate units are recalcitrant to ring-opening, while its labile glycol units are prone to unproductive decomposition into volatile C_1 species. Thus, the structural 'simplicity' of PE grants it a higher overall carbon efficiency in the two-step conversion to acetate, explaining the observed yield difference. Extending this structure-activity analysis to polyvinyl chloride (PVC) led to a paradigm-shifting discovery. In contrast to the recalcitrance imparted by the aromatic ring in PET, the C—Cl bonds in PVC, rather than inhibiting the reaction, actively promoted it, resulting in the highest degradation efficiency and acetate yield observed (Figure S21 and Table S1). We hypothesize that chloride ions (Cl^-) released during the initial degradation are converted into highly reactive chlorine radicals (*Cl) by photogenerated holes or *OH radicals. These *Cl radicals, with their high oxidation potential, then indiscriminately attack the polymer backbone, leading to an accelerated chain scission process that overcomes the inherent stability of the C—Cl bond and ultimately enhances the formation of acetate precursors.

To better explain the mechanism of the Fe@C_3N_4 SAC/H_2O_2/Vis system, we systematically studied several factors that may affect the photocatalytic activity. The combination of H_2O_2, catalysts and light is essential for the upcycling of PET, as the reaction fails to produce acetate in the absence of any of these components. Crucially, replacing H_2O_2 with an O_2 atmosphere also yielded no acetate (Figure 2d), indicating the insufficiency of in situ photocatalytic H_2O_2 generation. This is quantitatively supported by measurements showing only a trace amount of in situ photocatalytic H_2O_2 generation (≈7×10^{-4} M in 4 h) under O_2 using Fe@C_3N_4 SAC (Figure S22), which is vastly inadequate compared to the rapid consumption of added H_2O_2 required for efficient degradation (Figure S18). Thus, externally supplied H_2O_2 is the indispensable precursor for the *OH responsible for polymer cleavage. To identify the primary ROS generated during photooxidation processes, different quenchers (Tert-butanol (TBA), p-benzoquinone (p-BQ), Ethylenediaminetetraacetic acid disodium salt (EDTA-2Na)) were applied for trapping and quenching ROS species, specifically *OH radicals, superoxide radicals (O^{2-}) and hole (h^+), respectively (Figure 2d). It was observed that TBA (w/o *OH), p-BQ (w/o $O_2^{\cdot-}$) and EDTA-2Na (w/o h^+) all slowed down the photocatalytic upcycling processes of PET for the Fe@C_3N_4 SAC/H_2O_2/Vis system, with TBA (w/o *OH) having the most prominent inhibitory effect on the system (Figure 2d). These results show that photocatalytic upcycling processes are facilitated by *OH radicals, and confirm that *OH radicals are the main functional ROS free radicals. Subsequently, these functional ROS free radicals would attack the plastics, degrading them into small molecule products (Figure 2a). While H_2O_2 could facilitate the hydroxyl functionalization of graphitic carbon nitride, our results do not show evidence of functionalization-induced instability of the catalyst. This may be due to the generated *OH are used to oxidize plastics rather than to functionalize the catalyst. This is consistent with the unchanged ICP results before versus after reaction (0.5 wt.%), and XPS results (Figure S7). These findings collectively confirm that the catalyst maintains its structural and compositional integrity during the photocatalytic process.

The effect of additives is mentioned here.

To investigate the effect of additives in commercial plastics, and the effect of light trapping in the reactor, we systematically studied these two effects on the product yield of acetic acid from photocatalytic upcycling processes (Figure 2e–g and Figure S23). Comparing the catalytic reaction of commercial PET versus pure PET under the same conditions, the additives in commercial plastic lowered the yield of acetic acid (Figure 2e). This detrimental effect was further compounded in a post-consumer plastic mixture, which exhibited an even lower acetate yield than that of the pure plastic mixture (Figure S23). We attribute this to the synergistic negative effects of additives and the competitive kinetics between different polymer types during concurrent degradation. Processes to remove the additives from commercial plastics can therefore improve the yield of acetic acid. The yield of acetic acid was significantly enhanced by light trapping via wrapping the reactor with Al foil (Figure 2e–g), up to five times that of the unwrapped reactor (in the case of commercial PET). This improvement is attributed to the reflective properties of the Al foil, which reflects light back into the reaction medium, thereby increasing photon transport within the reactor. As a result, photons are effectively trapped and guided by internal reflection (Figure 2a), enhancing the overall light utilization efficiency. This reveals the importance of light trapping and photon transport in the reactor, which can increase the efficiency of light utilization by the reactor.

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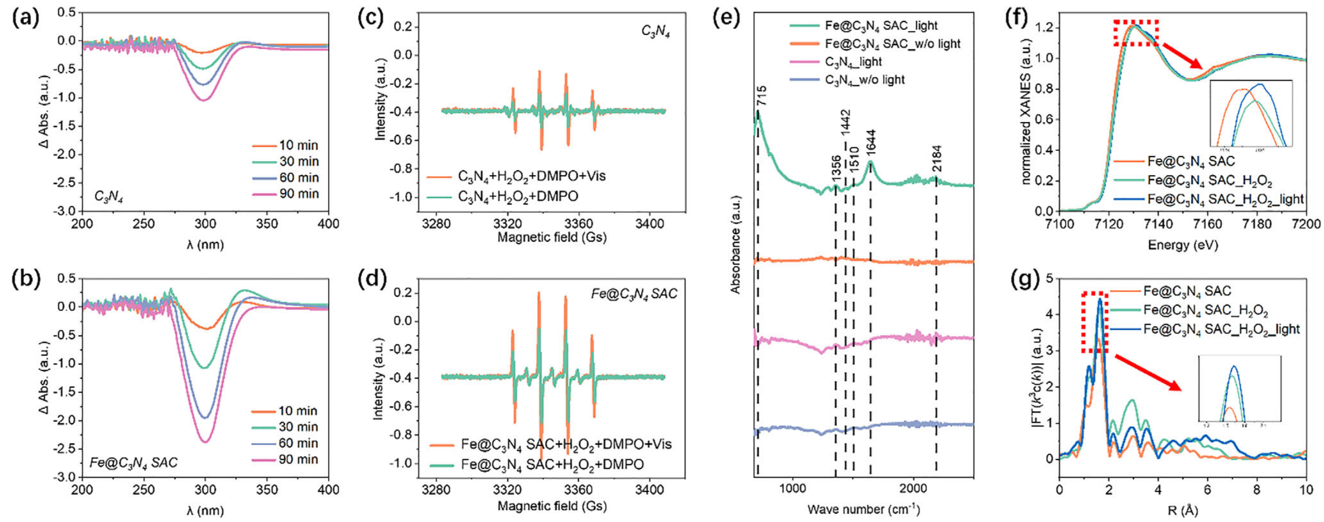

FIGURE 3 | *Operando* multimodal characterizations. *Operando* UV–vis spectra of (a) C_3N_4 and (b) $Fe@C_3N_4$ SAC during photocatalytic upcycling processes. *Operando* electron paramagnetic resonance spectroscopy (EPR) spectra of the DMPO-*OH over (c) C_3N_4 and (d) $Fe@C_3N_4$ SAC solutions with/without visible light irradiation. (e) *Operando* Fourier transform infrared spectroscopy (FTIR) spectra of C_3N_4 and $Fe@C_3N_4$ SAC. *Operando* synchrotron hard X-ray absorption spectroscopy of $Fe@C_3N_4$ SAC during the cascade photocatalysis from Hard X-ray MicroAnalysis (HXMA). (f) Fe K-edge XANES. (g) The corresponding EXAFS R-space results.

1.3 | In Situ Multimodal Characterization of the $Fe@C_3N_4$ Photocatalyst During Cascade Photocatalysis

To further identify the *OH radicals formed during the PET photocatalytic upcycling processes, ultraviolet-visible (UV-Vis) absorption spectra were obtained (Figure 3a,b). It is known that *OH radicals can be directly detected by UV-Vis through adding salicylic acid (SA) [56, 57]. According to the results, the $Fe@C_3N_4$ SAC system has a higher *OH generation ability compared to the C_3N_4 system under the same conditions (Figure 3a,b). To cross-verify, electron paramagnetic resonance (EPR) spectra were employed (Figure 3c,d). Apparently, characteristic peaks of 1:2:2:1 ratio existed in all systems (Figure 3c,d), which could be assigned to DMPO-*OH [58]. Importantly, for a given catalyst (C_3N_4 or $Fe@C_3N_4$ SAC), the *OH EPR signal intensity increased significantly under light irradiation, attributed to the accelerated decomposition of H_2O_2 into *OH with the help of photocatalyst. While the peak positions remained identical, the signals for $Fe@C_3N_4$ SAC were markedly stronger than those for C_3N_4, indicating a higher *OH generation capability by the $Fe@C_3N_4$ SAC during the photocatalytic upcycling processes (Figure 3c,d). Therefore, all analyses above revealed that the efficiency of photocatalytic plastic upcycling is enhanced by the introduction of Fe single atoms as active sites.

To detect the intermediate species and products during the photocatalytic plastic upcycling processes, *operando* Fourier transform infrared spectroscopy (FTIR) was performed. The vibrational peaks for the $Fe@C_3N_4$ SAC system are more pronounced than C_3N_4 (Figure 3e). Please note that the infrared beam was focused on the liquid layer. After illumination, several new peaks emerged, suggesting the formation of reactive intermediates and acetic acid as the dominant liquid-phase product. The band at 1644 cm^{-1} corresponds to H–O–H bending of adsorbed water [59], which overlaps with the expected C═O stretching of acetic acid [60]. Nonetheless, the formation of acetic acid was unequivocally confirmed by the presence of the following characteristic peaks: the acetate peak at 715 cm^{-1} (bending vibration of the C═O bond in CH_3COO^-), the symmetric COO^- stretching vibration at 1442 cm^{-1}, and the asymmetric COO^- stretching vibration at 1510 cm^{-1} [61, 62]. The intensity of these peaks increases, indicating that *COOH acts as an intermediate in the production of acetic acid [63, 64]. This is also confirmed by DFT calculations (Figure 4). The band at 1356 cm^{-1} is consistent with the CH_3 bending modes of acetate species [60], though contributions from surface carbonates cannot be entirely ruled out [65]. Importantly, a distinct peak emerged at 2184 cm^{-1}, which is tentatively assigned to the stretching vibration of linearly adsorbed carbon monoxide (CO) on metal sites [65]. This suggests that CO is a transient intermediate, potentially arising from decarboxylation or C─C bond cleavage steps during the oxidation process. Although CO_2 was not detected directly in the liquid phase, its formation as a short-lived intermediate is inferred from the selective oxidation of PET and the exclusive production of acetic acid.

To further reveal the valence state and local structure of Fe, the *operando* synchrotron X-ray absorption spectra were analyzed. Figure 3f shows in situ XANES Fe K-edge of $Fe@C_3N_4$ SAC under reaction conditions. Apparently, the changes of absorption edge are not obvious, suggesting that the Fe valence state is stable before and after light irradiation, which is also confirmed by DFT calculations (in the inset of Figure 4g). One can observe that the average valence of Fe is +1.1, with a relatively narrow distribution from +0.9 to +1.8. Though a high Fe valence is observed at the *COOHCH intermediate, as shown in Figure 4g, the intermediates of (*CO)(*CHO) and *COCHO are more stable than the *COOHCH intermediate (Figure 4f). Thus, a metastable *COOHCH intermediate with high Fe valence shows a short duration. As shown by the magnified white line peak (the inset of Figure 3f), the peak intensity increases under light

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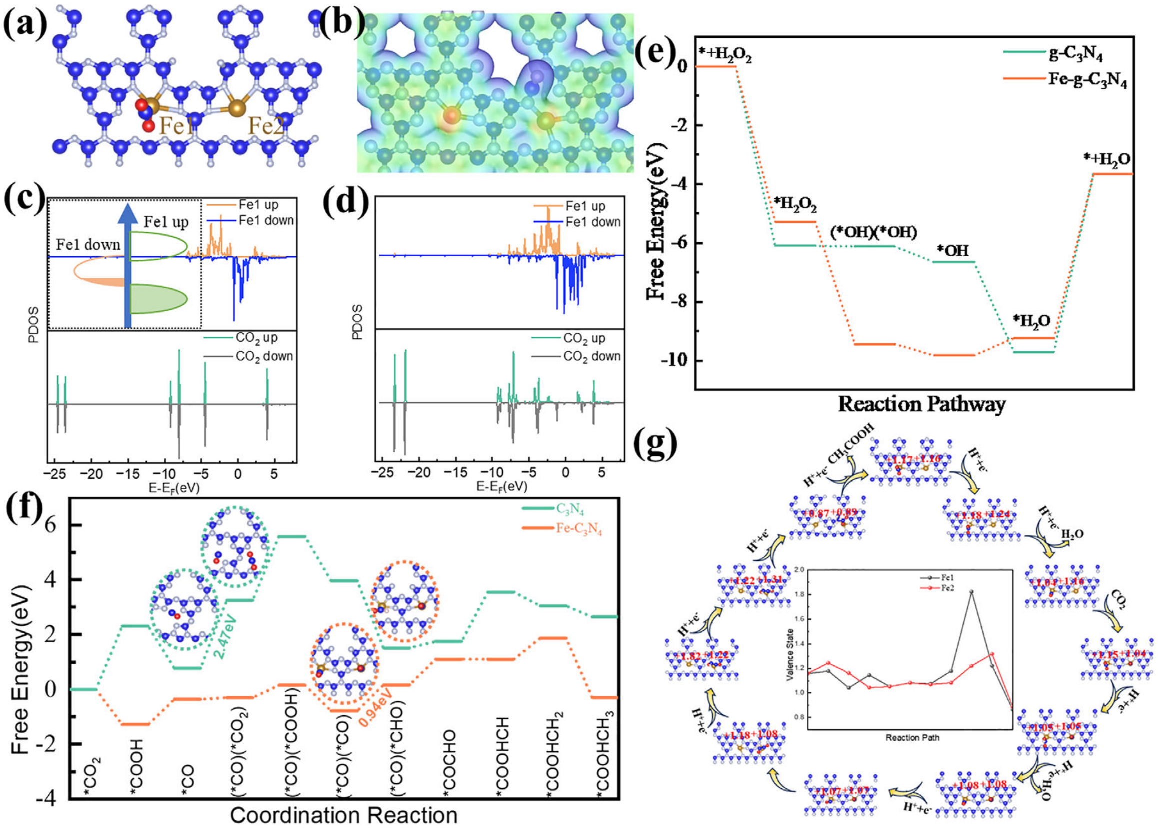

FIGURE 4 | DFT calculation of reaction pathways on Fe@C_3N_4 SAC and C_3N_4 photocatalysts via cascade photocatalysis. (a) atomic configurations of Fe@C_3N_4 SAC and (b) its electronic potential distribution. Projected density of states (PDOS) of Fe and CO_2 (c) before and (d) after the adsorption of CO_2 on the Fe1 site of Fe@C_3N_4 SAC. (e) Calculated Gibbs free energy diagram of the Fenton reaction pathway. (f) Calculated Gibbs free energy diagram of the CO_2 reduction reaction pathway for Fe@C_3N_4 SAC and C_3N_4 with (g) their reaction pathway. Fe: golden atom, C: blue atom, N: silver atom, O: red atom.

irradiation, indicating an occurrence of electron transfer [66, 67]. In addition, the local coordination environment of Fe was studied by comparing the EXAFS of Fe K-edge spectra (Figure 3g). There is a slight increase in the intensity of the Fe-N peak, which elucidates the alternation of the local Fe atomic arrangement [66]. The corresponding fitting curves are shown in Figure S24 and S25 and the fitting parameters are provided in Table S4. We observed that the coordination number changed from ≈4 to ≈5 and then back to ≈4 during the cascade photocatalysis, which is also confirmed by the DFT calculations (Figure 4a). The Fe─N bond length is as high as ≈2.02 Å, in agreement with the fitting results for synchrotron soft X-ray spectroscopy characterization in Table S4.

1.4 | Density Functional Theory Calculation (DFT) of Reaction Pathways

DFT calculations were performed to determine the role of Fe and carbon vacancy on acetic acid production based on the photocatalysis of Fe@C_3N_4 SAC. Detailed information was provided (Experimental Section). The Raman spectra (Figure 1i) confirmed the existence of defects in the C_3N_4 and Fe@C_3N_4 SAC (Figure 4a). Thus, we first evaluated the carbon defect energy of C_3N_4 for a thermodynamically stable structure (Figure S26). The defect energy for carbon vacancy was 1.7 and 2.8 eV, respectively, for C_3N_4 and Fe@C_3N_4 SAC in thermodynamically stable structures (Figure S26). Further, we determined the adsorption energy of CO_2 on Fe@C_3N_4 SAC and C_3N_4 to be −0.7 and −3.5 eV, respectively (Figure S27). The strong adsorption of CO_2 on C_3N_4 was not favorable for the further transformation of CO_2 to *COOH. Based on the analysis of projected density of states (PDOS) (Figure S28), CO_2 bands were seldom changed before and after its adsorption on C_3N_4 (Figure S28c,d), revealing that the Van der Waals force was vital in facilitating the adsorption of CO_2 on carbon-vacancy C_3N_4, rather than strong chemical bonds. In sharp contrast, Fe single atom sites in Fe@C_3N_4 SAC were the electron center, beneficial to adsorb CO_2 (Figure 4b) and form strong chemical bonds for the activation of CO_2. After CO_2 adsorption on Fe@C_3N_4 SAC, the energy level of Fe from -10 to +4 eV overlapped with that of CO_2 (Figure 4d), meaning Fe1 has an interaction with CO_2 via robust chemical bonds. In comparison with that before adsorption (Figure 4c,d), the energy level of Fe1 slightly descended and the counterpart of CO_2 rose up, revealing that an electron was transferred from Fe1 to CO_2 and thus CO_2 was activated. Due to the high spin state of Fe in Fe@C_3N_4 SAC, the spin-up Fe shows a non-negligible gap between its energy peak and the Fermi level, thus posing a difficulty in transferring electrons between Fe sites and CO_2 (mainly energy overlap). Though spin-down Fe shows a negligible energy overlap with CO_2 (low interaction), it crosses the Fermi level, thus facilitating electron transfer from the

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spin-down to spin-up state (in the inset of Figure 4c). In other words, the asymmetrical distribution of electrons, especially near the Fermi level, lowers the electron scattering from spin-up and spin-down states, thus accelerating electron transfer and promoting the fast transformation of CO_2 into other key intermediates. In contrast, C_3N_4 needs extra energy to proceed to other key intermediates (Figure 4f; Figure S28c,d). Further, a weaker adsorption energy of CO_2 on Fe@C_3N_4 SAC effectively promoted the transformation of CO_2 into other key intermediates. In addition, the cascade photocatalysis reaction pathway of the Fenton reaction and CO_2 reduction reaction to acetic acid was calculated. Based on the proposed Fenton reaction pathway (Figure 4e; Figure S29), it was found that the free energy of *OH was the lowest among the intermediates on Fe@C_3N_4 SAC, thus leading to increased accumulation of *OH on Fe@C_3N_4 SAC. An estimated barrier of ≈0.6 eV was needed for the transformation of *OH into *H_2O. As expected, these accumulated *OH radical species promoted plastics oxidation into CO_2 intermediates on Fe@C_3N_4 SAC. In comparison, the most stable structure among the proposed radical structures on C_3N_4 was *H_2O, leading to increased consumption of adsorbed *OH. We also simulated the reaction pathway of CO_2 intermediates to acetic acid on Fe@C_3N_4 SAC and C_3N_4. Regardless of the role of Fenton reaction on the CO_2 reduction reaction, the estimated barriers were 0.94 and 2.47 eV, respectively, for Fe@C_3N_4 SAC and C_3N_4 (in Figure 4f,g; Figures S30 and S31). The rate-determined step on Fe@C_3N_4 SAC was the transformation of (*CO)(*CO) into (*CO)(*CHO) (Figure 4f). By contrast, CO_2 adsorption on C_3N_4 due to the Van der Waals force was not favorable for further CO_2 adsorption, and thus the rate-determined step on C_3N_4 was the transformation of *CO into (*CO)(*CO_2) (Figure 4f). Hence, in this study, the Fe single atoms and carbon vacancy were indeed the key to modulate the local electronic structure for cascade Fenton and CO_2 reduction reactions on the same catalyst. Besides, to demonstrate the cascade effect of the catalyst, free energy diagrams for the Fenton reaction were compared. On C_3N_4, the adsorption of H_2O forms the most stable structure, and starting from this C_3N_4-H_2O configuration, the calculated energy barrier for reducing CO_2 to CH_3COOH is 1.72 eV (Figure S32), comparing with the barrier of 2.64 eV without water adsorption on C_3N_4 (Figure S31). This confirms that water adsorption facilitates the CO_2 reduction process compared to the pathway without water adsorption (Figure S31). Similarly, for the Fe@C_3N_4 SAC in the Fenton reaction, OH adsorption leads to the most stable structure, and along this pathway, the energy barrier for catalytically reducing CO_2 to CH_3COOH is significantly lower at only 0.94 eV (Figure S32), comparing with 1.22 eV without hydroxyl radical (Figure S31). These findings collectively support the feasibility of the cascade reaction occurring on the same photocatalyst.

1.5 | Techno-Economic Analysis

To assess and inform the commercial profitability of Fe@C_3N_4 SAC for large-scale plastics degradation, a techno-economic analysis was performed. The net benefits of converting PET into acetic acid can be calculated by identifying the costs of the necessary material, energy and labor inputs (Table S5 and Note S1). Table S5 summarizes the hourly costs and benefits of using a 400, 000-fold scaled-up reactor (the actual volume used is 12 m^3), where the average 2023 prices for the North American market for the inputs and outputs are derived by averaging available chemical and other input data from various official sources over the past three years. Only the variable costs are presented. The required inputs are based on the lab experiments presented in this paper and would need further calibration and validation in a real-world setting for future upscaling, accounting for possible economies of scale. In a real-world setting, there would be fewer labor operations needed in the reaction process, and the hourly manual solution addition could be replaced by an auto-sampling machine. The amount of labor and, hence labor costs to operate the conversion process would furthermore be largely independent of the number of reactors. Also, compared to plastics incineration, the amount of energy required for upgrading is negligible ($<1\%$) [68].

The financial profitability analysis shows that the costs are entirely (99.9%) driven by the high amount of hydrogen peroxide needed in the plastic waste upgrading process, yielding a net loss. However, the market price of acetic acid does not account for the broader social and environmental benefits of reducing or avoiding plastic pollution. Currently, plastic waste mainly ends up in the environment or in landfills (79%), where they degrade over very long periods of time (hundreds of years). Only a small share is recycled (9%) or incinerated (12%) [1]. Incineration requires significant energy and labor costs that would be avoided, as well as the emission of CO_2 [69]. The proposed plastics upgrading process using cascade photocatalysis would avoid these operation costs for a small share of the total amount of plastic waste and, more importantly, the adverse environmental effects of almost 80% of all plastic ending up as litter in the environment. Accounting for the social costs of plastic pollution [70] yields a total benefit that is more than twice as high as the estimated variable costs, and expected to be able to cover both the fixed (one-off) material costs presented in Table S6, which did not degrade during the experiment, and the fixed capital costs of the necessary reactors and installation space to scale up the lab process to industry level. These total fixed costs cannot be higher than $580 thousand per tonne to break even. This is unlikely if we compare the fixed capital costs for plastics upgrading, for example, with those for landfills that require a lot more space [71]. Scaling up to industry level will require substantial financial government support, for example through subsidies or other incentives like tax incentives or grants, to capture the broader societal benefits of plastics recycling. Please note that with the advancements in electrochemical synthesis, the price of H_2O_2 production keeps decreasing. Ideally, in situ generation of H_2O_2 by renewable electricity could be employed as feedstock to the reaction. This would further decrease the cost of this process.

2 | Conclusion

In summary, a promising strategy to upcycling plastics via photocatalysis was demonstrated and excellent production rates of acetic acid were observed. Plastic products were converted into CO_2 as intermediates first, then successfully converted into acetic acid on the same photocatalyst via cascade reactions without the addition of other sacrifices. In conclusion, this work not only demonstrates a photocatalytic process for remediating plastic wastes in a simulated natural and real sunlight environment,

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but also shows the possibilities of single-atom catalysts in plastic upcycling. Looking forward, the efficiency and sustainability of such photocatalytic upcycling systems can be further enhanced through strategic engineering at multiple levels. At the reactor design level, the application of photon transport and light trapping strategies can maximize light utilization. Concurrently, at the process integration level, the efficiency could be substantially improved by coupling with a more sustainable oxidant supply, such as in situ electrocatalytic H_2O_2 generation, which could feed directly into the upcycling reactor to create a more integrated and potentially self-sustaining process.

3 | Experimental Section

3.1 | Materials

Melamine ($C_3H_6N_6$, 99%), iron (III) chloride hexahydrate ($FeCl_3{\cdot}6H_2O$, ACS reagent, 97%), hydrogen peroxide solution (H_2O_2, 30% (w/w) in H_2O, contains stabilizer), polyethylene terephthalate (PET, size: 0.25 cm × 0.35 cm × 0.35 cm), polypropylene (PP, isotactic, average Mw ≈12, 000, average Mn ≈5, 000, size: 0.6 cm × 0.7 cm × 0.25 cm) and polyethylene (PE, average Mw ≈4, 000 by GPC, average Mn ≈1, 700 by GPC, size: between 5 um and 800 um) were purchased from Sigma–Aldrich. All chemical reagents were used without further purification.

3.2 | Synthesis of Graphitic Carbon Nitride (C_3N_4)

A certain amount of melamine was directly heated to 550°C for 4 h in argon to obtain the final graphitic carbon nitride (denoted as C_3N_4) photocatalysts.

3.3 | Synthesis of Fe@C_3N_4 SAC

Fe@C_3N_4 SAC catalysts were prepared in a typical procedure from a previous study [72]. A homogeneous solution of 2 g melamine and 20 mg $FeCl_3{\cdot}6H_2O$ was dispersed separately in ultrapure water and then mixed together. After continuous magnetic stirring at 100°C, the water in the resulting mixture was completely evaporated. Afterwards, the obtained sample was calcined in a tube furnace at 550°C for 4 h under an argon atmosphere. The as-prepared sample was then collected and labeled as Fe@C_3N_4 SAC. The mass ratio of iron (0.5 wt.%) in Fe@C_3N_4 SAC was determined by inductively coupled plasma mass spectrometry (ICP-MS).

3.4 | Material Characterization

Scanning electron microscopy (SEM) images were tested by a Hitachi S4800 with a working accelerating voltage of 10 kV. Glancing-incidence X-ray diffraction (GIXRD) results were collected using a PANalytical X'Pert Pro MRD diffractometer with Cu Kα radiation (1.54 Å) at an incidence angle of 1°. Brunauer–Emmett–Teller (BET) was gained on a Micromeritics Gemini VII 2390a. X-ray photoelectron spectroscopy (XPS) measurements were acquired from a Thermo-VG Scientific ESCALab 250 microprobe with a monochromatic Al Kα X-ray source (1486.6 eV). The obtained spectra were calibrated using C 1s line. Probe aberration-corrected scanning transmission electron microscope (AC-STEM) tests were carried out on a JEOL ARM200CF TEM equipped with energy-dispersive X-ray spectroscopy (EDS) at 200 kV. The HAADF-STEM images were recorded by JEOL ARM200CF STEM with a probe aberration corrector operating at 200 kV. Inductively coupled plasma mass spectrometry (ICP-MS) analyses for Fe content were carried out on an Agilent 8800 triple quadrupole ICP-MS, using He as a collision cell gas and Sc as an internal standard to correct for instrument drift. The primary ICP calibration standard was from Aristar VWR Chemicals BDH. A secondary standard from Delta Scientific Inorganic Ventures was used to confirm instrument accuracy (within 7%; relative standard deviation for individual sample analyses was ≤1.1%). Raman spectroscopy was collected by a Bruker SENTERRA Dispersive Raman Microscope to analyze the degree of graphitization and defects of the sample using the excitation wavelength of 785 nm. The laser beam was focused onto the sample's surface by an L×50 objective. The aperture is 50×1000 um and the resolution is 3–5 cm^{-1}. Each spectrum was integrated over 0.01 s at 50 mW laser power. At least ten spectra in the range of 40–1520 cm^{-1} were collected from randomly selected points on the surface of each sample.

3.5 | Photocatalytic Oxidation of Plastics

The photocatalytic activities of catalysts were mainly evaluated by the upcycling of different plastics (i.e., polyethylene terephthalate (PET), polypropylene (PP) and polyethylene (PE), commodity PET from DuPont with 30% glass fibers) in a simulated natural environment with AM1.5G irradiation from a solar simulator. In a typical experiment, a certain amount of catalyst was dispersed in deionized water. After stirring well for several hours, the sediment was removed by centrifugation. Then 30 mL solution was added into the reactor. The reaction pH, initially adjusted to 3 with H_2SO_4, was qualitatively confirmed to remain stable throughout the process, as no visible change was observed using pH test strips, consistent with the low concentration of weak acidic products formed. The catalyst usage is 8 mg unless otherwise stated. Afterwards, the well-dispersed solution was transferred into the reactor and 2 mL of H_2O_2 was added. Before light irradiation, the reactor was shielded to the light, ensuring that the adsorption equilibrium was established. During the photoreaction processes, the reaction system was under continuous magnetic stirring, while the temperatures of the reactor were controlled at 298 ± 0.2 K by a recirculating cooling water system during irradiation. The light source for the photocatalysis was a 300 W Xe lamp (PLS-SXE300DUV, Beijing PerfectLight Technology Co., Ltd, China) with a standard AM 1.5G filter, which could provide the one-sun irradiation from the top of the reactor. In order to investigate the strategy to enhance the photon transport and utilization efficiency, we sealed the reactor by wrapping the outer wall of the reactor with Al foil. The conversion rate was measured by adding 2 mL H_2O_2 into catalysts and plastic solution every 3 h. The weight of PET has been obtained before light irradiation. After long time exposure, PET was dried and then was weighed to calculate the weight loss.

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3.6 | Photocatalytic Oxidation Under Real Sunlight Illumination

The photocatalytic oxidation was also performed under natural sunlight (≈0.6 sun) to evaluate the practical feasibility of the process. A convex lens was used to focus direct sunlight onto the reaction vessel, maintaining an illuminated spot area and resultant light intensity comparable to the standard laboratory setup (Figure S14). All other experimental parameters, including catalyst loading, plastic mass, and solution volume, were kept identical to the controlled laboratory experiments. The reaction was initiated only after the addition of 2 mL of H_2O_2 to the system. Samples were periodically withdrawn from the reaction mixture at 1 h intervals for subsequent H-NMR analysis.

3.7 | H-NMR Spectroscopy Measurement

Proton nuclear magnetic resonance (H-NMR) was measured on a Bruker Avance III 300 MHz. A standard curve of acetic acid was first built using pure chemicals with known concentrations. NMR testing samples were prepared by combining 80 µL of aqueous 0.5 mg mL^{-1} maleic acid internal standard with 600 uL sample. After photocatalytic reaction, the solution was collected and mixed with the internal standard solution. The solutions are mixed in the same proportions as above. For H-NMR tests, 128 scans were performed, with excitation sculpting used to suppress the water peak. As presented in Figure S34a, acetate and maleic acid can be ascribed to the peaks located at 1.8 (triplet), and 6.1 ppm, respectively.

The concentration of liquid products could be calculated by the following equation (Figure S34b):

$$\text{Area Ratio to Maleic Acid} = 3.76 \times \text{Concentration (mM)} + 0.04$$

3.8 | *Operando* UV–Vis Spectroscopy Measurement

Irradiation of well-dispersed catalysts solution and salicylic acid (SA) mixtures was performed in the reactor. The solution was collected at certain time intervals (minutes) and the concentration of *OH radicals was determined from UV–Vis spectra acquired between 200 nm and 800 nm using a Shimadzu UV-2600i spectrophotometer [56, 57].

3.9 | *Operando* Electron Paramagnetic Resonance (EPR) Spectroscopy Measurement

In situ EPR spectra were recorded on an EPR spectrometer (Bruker EMX-Micro X-Band) at room temperature. Hydroxyl radicals (*OH) were captured using 5, 5-dimethyl-1-pyrroline N-oxide (DMPO). In typical measurements, a well-dispersed catalysts solution was prepared followed by the addition of DMPO. The concentration of DMPO is 0.14 M in the final solution. EPR spectra were recorded before the irradiation. After 10 min irradiation on the solution in simulated natural environments under AM1.5G irradiation, the solution was analyzed a second time.

3.10 | Ex situ Synchrotron Based X-Ray Absorption Spectroscopy (XAS)

Ex situ hard XAS measurements were carried out at the 20-BM and 20-ID-C beamline of the Advanced Photon Source (APS), Argonne National Laboratory. The measurements at the Fe K-edge were performed in fluorescence mode using a Lytle detector. Ex situ Fe K-edge XAS measurements were also carried out in fluorescence mode at soft X-ray microcharacterization beamline (SXRMB) at the Canadian Light Source. The XAS data fitting for the first coordination shell was performed using the Artemis implemented in the FEFF software packages. The k and R ranges for fitting results were k = 0–12 $Å^{-1}$ and R = 1–3 Å, respectively and were used throughout the study. Values for the amplitude reduction factor was determined to be 0.8 from the Fe foil reference spectra, which was applied to the whole analysis. Fitting parameters were the coordination numbers (CN), the inner potential correction (ΔE), Debye–Waller factors (σ [2]), atomic distances (R-factor). The quality of fitting results was assessed by the R-factor. R-factor Numbers less than or equal to 0.02 are considered acceptable [73].

3.11 | *Operando* XAS Measurements

The measurements were conducted at 06ID-1 Hard X-ray Micro-Analysis (HXMA) beamline of the Canadian Light Source (CLS), operated at 2.9 GeV with a constant current of 250 mA. The *operando* measurements at the Fe K-edge were performed in fluorescence mode using a Lytle detector. The XAS data fitting for the first coordination shell was performed using the Artemis implemented in the FEFF software packages.

3.12 | *Operando* FTIR Measurements

The measurements were conducted at 02B1-1 Far-IR beamline of the Canadian Light Source (CLS). Using a Bruker IFS125HR spectrometer equipped with a modified Pike Instruments Glad-iATR diamond Attenuated Total Reflectance system. Setup of in situ reaction is shown in Figure S35. The 2 mL of a 1 mg mL^{-1} catalyst solution, 20 mg of PET and H_2O_2 was placed on the ATR diamond. Subsequently, the mixture was pressed by quartz window to ensure good contact between the ATR diamond and the catalysts/plastic powder. The light source was from luzchem (model: LUZ-XE-PV S/N). The data prior to illumination and the data collected every 20 min post-illumination were gathered.

3.13 | DFT Computational Details

Density functional theory (DFT) calculations were performed in the Vienna ab initio simulation package (VASP). The generalized gradient approximation with the Perdew-Burke-Emzerhof parameterization was used to approximate the exchange-correlation functional. The electron-ion interactions were described by the projector augmented wave (PAW) method. A plane-wave basis with the cutoff energy of 400 eV was used. The Brillouin zone was built using a $1 \times 2 \times 1$ Gamma grid. The geometries got convergence until the energy of 1.0×10^{-5} eV/atom and the force of 0.05 eV $Å^{-1}$. Van der Waals with DFT-D3 was

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considered in these simulations. A vacuum layer of 15 Å along the z direction was constructed to avoid periodic interaction.

To determine the carbon defect sites, the defect energy (E_d) was calculated as the following:

$$E_d = E_{defect} + E_C - E_{prefect}$$

wherein, E_{defect} was the energy of defect model, E_c was the energy of single carbon atom from bulk carbon, $E_{prefect}$ was the energy of ideal model. A lower defect energy corresponds to higher stability.

To compare the adsorption of CO_2 on the substrates, the adsorption energy (E_{ads}) was evaluated as follows:

$$E_{ads} = E_{total} - E_{gas} - E_{sub}$$

wherein, E_{total} was the energy of gas adsorbed on the substrates, E_{gas} was the energy of CO_2, E_{sub} was the energy of the substrates. A more negative adsorption energy meant a more stable adsorption.

Gibbs free energy, presented by Nørskov et al. [74], was calculated according to the following equation:

$$\Delta G = \Delta E + \Delta ZPE - T\Delta S$$

wherein, ΔE was the electronic adsorption energy, ΔZPE was the zero point energy difference between the adsorption and gaseous species, and $T\Delta S$ was the corresponding entropy difference between these two states. The free energy of H_2O in bulk water was calculated under the equilibrium vapor pressure (0.035 bar and 298.15K). The free energy of $(H^+ + e^-)$ was evaluated as the energy of 1/2 H_2 at standard conditions (1 bar and 298.15K).

Author Contributions

Y.A.W. conceived and supervised the project. W.W. and C.D. carried out the catalyst's synthesis, characterization, and performance test. J.G. conducted the DFT calculations. X.W., M.W., H.Z., C.J.S. and M.S. carried out the ex situ XAFS measurements and data fitting. W. W., X. W., Z. C., M.Z., and L. W. perform in situ FTIR, in situ XAS. W.W., P.S., and F.F.S. performed in situ EPR measurement. N.C. and W.C., assisted with in situ XAS measurement. B.B. assisted with in situ FTIR measurement. B.K. performed ICP test. R.B. assisted with the techno-economic analysis. W.W., C.D., J.G., and Y.A.W. wrote the manuscript. All authors made comments and helped refine the manuscript.

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Conflicts of Interest

The authors declare the following conflict of interest: Y.A.W., W.W., and C.D. have filed a US patent application (No. 19/089933) on this technology related to this Photocatalytic upcycling waste micro-plastics into value-added products. The remaining authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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